Lab 08: Query Processing and Optimizer

Use lab11 folder

- `Change directory into **lab11** directory`
- `Save the file and extract it under Lab folder`
- `Remember to start by creating a sanjose database to use.`

A. Query Processing and Optimizer in MySQL

A1. Create sanjose Database with mysql

For this exercise lab, create sanjose database using mysql. Then, create COMPANY database using `cr_company.sql` file. If needed, you can use `lab1.sql` file.

```
mysql> CREATE DATABASE sanjose;
mysql> USE sanjose;
mysql> source cr_company.sql
```

After, you create COMPANY database on sanjose, you can check the tables using desc command.

A2. Check Catalog Database

MySQL catalog database is "INFORMATION_SCHEMA", and there exist many tables that store meta data in the mysql. Let's check several important tables: STATISTICS, TABLES, TABLE_CONSTRAINTS . . . if needed, you can use `lab2.sql` file.

```
mysql> use information_schema;
mysql> show tables;
mysql> desc tables;
mysql> desc statistics;
mysql> select table_name, table_rows, avg_row_length, data_length
      -->    from information_schema.tables
      -->  where table_name = 'EMPLOYEE';
mysql> select * from information_schema.statistics;
```

A3. Collect Statistical Data using ANALYZE

It is very important to keep the up to date information about the database in order to get accurate decision of query optimizer. MySQL uses `ANAYZE` command to collect statistical data. You can use `lab3.sql` file if needed.

First, delete one row (i.e., ssn = '333445555'). Then, check the catalog database if the deletion is applied or not:

```
mysql> use sanjose;
mysql> delete from employee where ssn = "333445555";
mysql> select table_name, table_rows, avg_row_length, data_length
      -->    from information_schema.tables
      -->   where table_name = 'EMPLOYEE';
```

Any Problem?
Run analyze table on "EMPLOYEE" to update the information about EMPLOYEE table:

```
mysql> analyze table employee;
mysql> select table_name, table_rows, avg_row_length, data_length
      -->    from information_schema.tables
      -->   where table_name = 'EMPLOYEE';
```

You can see the updated information about the table.

A4. Execution Plan

Execution plan is the output of query processor in MySQL, and it shows how MySQL accesses the database for a given query. In order to get the execution plan in MySQL, you can use EXPLAIN command as below: You can use lab4.sql file if needed.

- Simple Example: Check what is the plan of accessing employee table.
  ```
  mysql> explain select fname, lname from employee;
  mysql> explain select ssn from employee;
  ```

- Let's create Primary Key of EMPLOYEE on SSN
  ```
  mysql> alter table employee add primary key (ssn);
  mysql> analyze table employee;
  ```

- Check the execution plan again for the same query. What is the difference?
  ```
  mysql> explain select fname, lname from employee;
  mysql> explain select ssn from employee;
  ```

A5. More Complex Query

You can check the execution plan for more complex query. For example,

```
select d.dname, sum(e.salary)
from employee e, department d
where e.dno = d.dnumber
and e.salary >= 3000
group by d.dname;
```

Find the execution plan for above query. If needed, you can use `lab5.sql` file.

```
mysql> explain select d.dname, sum(e.salary)
       -->  from employee e, department d
       -->  where e.dno = d.dnumber
       -->  and e.salary >= 3000
       -->  group by d.dname;
```

Now, you can add Primary Key of DEPARTMENT on DNUMBER. Then, check the execution plan again.

```
mysql> alter table department add primary key (dnumber);
```

What is the difference?

Then, Remove e.salary >=, then check the execution plan again. Any difference you find? If so, why?

A6. Hint

In MySQL, you can force the query processor to use specific plan using HINT. For example, you can ignore to use PRIMARY KEY as below:

```
mysql> explain select ssn from employee;
mysql> explain select ssn from employee ignore index (PRIMARY);
```

Also, you can use 'USE INDEX(TYPE)' to specify an index to be used. You can find more examples of hint from the lecture slide.

Exercise: Query Optimization using Explain and Hint

First, find (or create) 3 SQL queries on COMPANY database including:

- `Q1.sql: Simple query (accessing single table with where)`
- `Q2.sql: Subquery (accessing at least 2 tables)`
- `Q3.sql: Aggerate query including Group By`

Second, check execution plan for each query

Third, you can try different query execution plan using HINT, then compare with the results from Second.